Dr. RADHE SHYAM SHARMA

Assistant Professor
Centre for Artificial Intelligence and Robotics (CAIR)
IIT Mandi, Kamand, Himachal Pradesh - 175005, India
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HIGHER ACADEMIC DETAILS

- **PhD** (Electrical Engineering)
  Institute: Indian Institute of Technology (IIT) Kanpur, India, 2020
  CPI: 9 out of 10

- **M.Tech.** (Modelling & Simulation)
  Institute: Defence Institute of Advanced Technology (DIAT) Pune, India, 2013
  Percentage: 75.25

RESEARCH INTERESTS

- `Visual Servoing`
- `Robot Learning`
- `Autonomous Navigation`
- `Multi-Robot Systems`
- `Digital Twin Technology`

PATENTS AND PUBLICATIONS

1. Yogesh Dasgaonkar, **Radhe Shyam Sharma**, “"A Method and System for Detecting and Segmenting Internal Impurities in an Object of Interest,” *IN Patent No. 567976*, 2025.
2. Yogesh Dasgaonkar, **Radhe Shyam Sharma**, “"System for Real Time Human Detection and Counting in a Closed Room Environment and a Method Thereof,” *IN Patent No. 549539*, 2024.
3. Deokrishna Kumar Choudhary, Yogesh Dasgaonkar, Dube Dheeraj Prakashchand, **Radhe Shyam Sharma**, Ravindra Naik Bukke “"Spectrophotometric Analysis Device, Method for Performing Spectrophotometric Analysis, and System and Method for Disease Prediction,” *IN Patent No. 587991*, 2026.
4. Tarun Sharma, Manikandan Ravikiran, Deepika Gupta, **Radhe Shyam Sharma**, Rohit Saluja, “Signal-to-Syntax: A Language-Based Framework for Few-Shot Classification Using Robot Manipulator With Haptic Perception,” *Accepted in IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)*, 2026.
5. Geeta Sharma, Sanjeev Jain, **Radhe Shyam Sharma**, “Sac-eprb: Soft Actor-Critic with Enhanced Prioritized Replay Buffer for UAV Navigation,” *Intelligent Service Robotics*, vol. 19, 2026.
6. Mayank Thakur, **Radhe Shyam Sharma**, Vinod Kurmi, Raj Samant, Badri Narayan Patro, “Spectral-Temporal Attention for Robust Change Detection,” *IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)*, 2025.

7. Geeta Sharma, Sanjeev Jain, **Radhe Shyam Sharma**, "Path Planning for Fully Autonomous UAVs-A Taxonomic Review and Future Perspectives," *IEEE Access*, 2025.

8. Geeta Sharma, Narinder Verma, Sanjeev Jain, **Radhe Shyam Sharma**, "A hybrid framework for secure IoT communication using lightweight cryptography and machine learning-based authentication," *Peer-to-Peer Networking and Applications*, vol. 17, 2025.

9. Yogesh Dasgaonkar, Mayank Thakur, Rahul, **Radhe Shyam Sharma**, "3D Perception for Safe-Gap-Aware Autonomous Drone Navigation," *IEEE International Conference on Emerging Technology in Autonomous Aerial Vehicles (ETAAV)*, 2025.

10. Rahul, Yogesh Dasgaonkar, Aachal Sharma, **Radhe Shyam Sharma**, Raj Samant, Narendra Kumar Dhar "Learning to Fly Together: Coordinated and Adaptive Drone Navigation Through Dynamic Movement Primitives," *IEEE International Conference on Robotics and Computer Vision (ICRCV)*, 2025.

11. Niharika Chaudhary, Yogesh Dasgaonkar, **Radhe Shyam Sharma**, Parimala Kancharla "MR-TEB: A Scalable Multi-Robot Trajectory Planning Framework with Adaptive Priority Scheduling," *IEEE International Conference on Automation, Robotics and Applications (ICARA)*, 2026.

12. Yogesh Dasgaonkar, **Radhe Shyam Sharma**, Raj Nayan Samant, "A Modular Training Approach for MultiRobot Cooperative Search in Dynamic Environments," *IEEE Indian Control Conference (ICC)*, 2024.

13. Yogesh Dasgaonkar, **Radhe Shyam Sharma**, "Through the Rubble: Discovering the Invisible," *IEEE International Conference on Control, Automation, Robotics and Vision (ICARCV)*, 2024.

14. **Radhe Shyam Sharma**, Arindam Mondal, Laxmidhar Behera, "Tracking Control of Mobile Robots in Formation in the Presence of Disturbances," *IEEE Transactions on Industrial Informatics*, vol. 17, no. 1, pp. 110–123, 2021. PDF Video of the Experiments

15. **Radhe Shyam Sharma**, Ranjith Ravindranathan Nair, Pooja Agrawal, Laxmidhar Behera, K. S. Venkatesh, "Robust Hybrid Visual Servoing using Reinforcement Learning and Finite Time Adaptive FOSMC," *IEEE Systems Journal*, vol. 13, no. 3, pp. 3467–3478, 2018. PDF Video of the Experiments

16. **Radhe Shyam Sharma**, Santosh Shukla, Laxmidhar Behera, Venkatesh K S, "Position Based Visual Servoing of a Mobile Robot With an Automatic Extrinsic Calibration Scheme," *Robotica*, vol. 38, no. 5, pp. 831–844, 2020. PDF Video of the Experiments

17. **Radhe Shyam Sharma**, Santosh Shukla, Hamad Karki, Amit Shukla, Laxmidhar Behera, Venkatesh K S, "DMP Based Trajectory Tracking for a Nonholonomic Mobile Robot With Automatic Goal Adaptation and Obstacle Avoidance," *IEEE International Conference on Robotics and Automation* (`ICRA`), pp. 8613-8619, 2019. PDF Video of the Experiments

18. Atulya Shivam Shree, **Radhe Shyam Sharma**, Laxmidhar Behera, K. S. Venkatesh, "Position Based Visual Control of The Hovering Quadcopter," International Conference on Intelligent Human Computer Interaction, pages 15–26. Springer, 2016. PDF Video of the Experiments

AWARDS AND FELLOWSHIPS

- Teaching Excellence Award
- Best Hostel Warden Award
- DAAD Postdoc-NeT-AI 4/2021 Fellow
- Travel grant by DAAD to visit several German Research Institutions
- Visiting research scholar at Khalifa University, Abu Dhabi, UAE, Oct - Nov 2018
- Received Institute fellowship for doctoral studies from Indian institute of Technology (IIT) Kanpur, India, Jan 2015 - Dec 2019
- Received GATE scholarship for Master of Technology course from MHRD, Govt of India, Aug 2011- May 2013

PROJECTS and POCs

- Capacity building for human resource development in Unmanned Aircraft System (Drone and related Technology)
  Ministry of Electronics and Information Technology, India
- Towards Industry 5.0 - Design Development of Drone Intelligence System to Accelerate Human-Robot Collaborative Learning for Heterogeneous Environment
- AeroPave: Autonomous Drone System for Pothole Detection in Low-Light and Noisy Environments
  Ministry of Electronics and Information Technology, India
- Agri-Eco Digital Twin: Drone-Assisted Real-Time Crop and Habitat Monitoring in Mountainous Regions
  Ministry of Electronics and Information Technology, India
- Digital Twin-Enabled Autonomous Firefighting Drone
  Ministry of Electronics and Information Technology, India
- Low-Latency Event-Driven Unauthorized Drone Identification for Defense and Aerospace Security
  Ministry of Electronics and Information Technology, India
- A condition monitoring system with multi-agent mechanism for external non-contact smart inspection of buried oil and gas pipelines
  The Petroleum Institute (PI), Khalifa University, Abu Dhabi, UAE
- Study on personal drone benefits for driver assistance
  Renault Nissan Technology and Business Center India (RNTBCI), Chennai, India
- Robotics and Automation in Agriculture
  Ministry of Electronics and Information Technology, India

INDUSTRIAL, RESEARCH AND TEACHING EXPERIENCE

- Assistant Professor
  Centre for Artificial Intelligence and Robotics (CAIR)
  IIT Mandi
  From 26-07-2022 to Till Date
- Autonomy Senior Engineer
  Integrated Components and Solutions Division
  Caterpillar
  From 05-07-2021 to 25-07-2022
- Project Post-Doctoral Fellow
  IIT Kanpur, India
  From 16-04-2021 to 30-06-2021
- Project Scientist
  IIT Kanpur, India
  From 16-06-2020 to 15-04-2021
- Research Associate
  IIT Kanpur, India
  From 01-01-2020 to 31-05-2020
- Faculty
  NIT Raipur, India
  From 15-07-2013 to 31-05-2014
- Lecturer
  SGSITS Indore, India
  From 15-07-2008 to 11-08-2011
- Lecturer
  SATI Vidisha, India
  From 11-09-2007 to 14-07-2008

COURSE TAUGHT

- AR511 - Autonomous Mobile Robots, IIT Mandi
- AR514 - Vision and Learning Based Control, IIT Mandi
- AR524 - Robot Simulators, IIT Mandi
- AR522 - Robot Vision, IIT Mandi
- IC102P - Foundation of Design Practicum, IIT Mandi
- AR504 - Robot Programming, Modeling and Simulation
- EE 301P - Control Systems Lab, IIT Mandi
- DP503P - Basic Mechanical and Electrical Workshop, IIT Mandi
- Tutor for Introduction to Electronics (ESc201A), 2018-19-II Semester, & 2019-20-I Semester, IIT Kanpur
- Tutor for Introduction to Electronics (ESc201A), 2019-20-II Semester, IIT Kanpur

TALKS DELIVERED

- Invited talk on "Drone Technology, Robotics and Autonomous Farm Machinery for Sustainable Agriculture", during a Workshop on Architecting a Strategic Roadmap for Scale-Based Adoption of Digital Agriculture for Viksit Bharat @2047 at RPCAU Pusa, Bihar, India, October 2026.
- Invited talk on "Human Skill Transfer to Aerial and Ground Robots", during a Workshop on AI-Enabled Drone Technologies for Societal and Industrial Impact at NIT Jalandhar, India, February 2026.
- Invited talk on "AI and Robotics in Agriculture" in a training program, Central Agricultural University, Pusa, Bihar, India, October 2025.
- Invited talk on "Robot Programming With ROS", FDP at MIT World Peace University, Pune, India, May 2025.
- Invited talk on "Robot Learning and Programming: From Theory to Real-World Applications",in a Workshop on Smart Automation: AI and Robotics for Industrial Needs, NIT Tiruchirapalli, India, November 2024.
- Invited talk on "Introduction to Robotics: AI Autonomous Systems and ROS Integration",in a Training Programme, SKUAST Kashmir, November 2024.
- Invited talk in a short-term course on "Recent Trends in Computer Vision, Artificial Intelligence & Robotics", at NIT Delhi, India, March 2023.
- Invited talk on "Smart Control Strategies for Autonomous Navigation of Mobile Robots" at TU Darmstadt, Germany, November 2022.
- Invited talk in a workshop on "Integration of Digital Technologies in Disaster Preparedness", at Kerala University of Digital Sciences, Innovation and Technology, Trivandrum, Kerala, India, May 2022.
- Invited talk in TEQIP -III Sponsored One Day Workshop on "Mathematics for Machine Learning", Coimbatore Institute of Technology, Coimbatore, India, March 2021.
- Invited talks (two sessions) on "Robot Operating System and its uses in robotics simulations" in the training program on "Future Skill training on Robotics and Automation", SGSITS Indore, India, September 2020.
- Invited talks (two sessions), AICTE Sponsored Two Week Faculty Development Programme on "Robotics and Autonomous Systems", PSG College of Technology, Coimbatore, November-December 2020.

PROFESSIONAL ASSOCIATIONS

- Reviewer for IEEE Robotics and Automation Letters
- Reviewer for IEEE Transactions on Mechatronics
- Reviewer for IEEE Systems Journal

- Reviewer for IEEE Transactions on Instrumentation and Measurement
- Reviewer for IEEE RO-MAN (Robot and Human Interactive Communication)
- Student member of IEEE
- Member of IEEE Robotics and Automation Society
- Visiting research scholar in the Department of Mechanical Engineering at Khalifa University, Abu Dhabi, UAE
- Speaker, GIAN (Global Initiative of Academic Networks) course on Selected Topics in Internet of Things, Jan 2018

PARTICIPATIONS IN CONSORTIUM/CONCLAVE/TRAINING PROGRAMS

- Drone Conclave, *Agricultural University, Palampur, July 4 - July 5, 2023*
- DRDO - Academia Conclave, *IIT Jodhpur, May 25 - May 26, 2023*
- Meity Consortium - National Programme on Perception and Cognitive Engineering, *Ministry of Defence, DRDO Bhavan, New Delhi, May 25 - May 26, 2023*
- Faculty Updation Program: - "Introduction to Navigation, Guidance and Control of Unmanned Systems", *IIT Bombay, March 13 - March 18, 2023*
- Training Program: - "Do Drone", *IISc Bengaluru, Nov 2022*
- Connected Vehicle 2022, Conference and Exhibition at *Bengaluru, May 2022*
- National workshop on "SVM in pattern recognition & its applications in biological sciences" at *NIT Raipur, Jan 2014*
- Faculty training/orientation program at *SGSITS Indore Dec 2008*
- Major training on "Imaging systems" at *AFMC* (`Armed Forces Medical College`) *Pune, Jun-Jul 2006*
- Minor training on "Biomedical Equipments" at *MH* (`Military Hospital`) *Gwalior, Jun-Jul 2005*

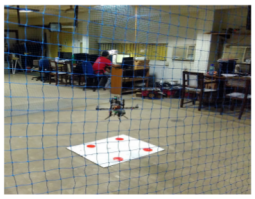
TECHNICAL SKILLS

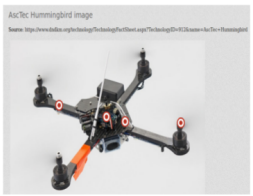
- Languages: MatLab, and Python
- Operating Systems: Linux, ROS (`Robot Operating System`)
- Others: PyBullet, LaTeX
- Hands-on experience with `unmanned ground` and `aerial vehicles` – Pioneer P3-DX, Guardian mobile platform, Segway RMP, Nayan, and AscTec Hummingbird.


(a) Pioneer P3-DX


(b) Guardian robot


(c) Segway RMP


(d) Nayan


(e) AscTec Hummingbird

TREKKING AND CAMPING EXPERIENCE

- Gaumukh Tapovan Trek
- Dodital Trek
- NSS CAMP

INTEREST AND HOBBIES

- Adventure trekking
- Robotics
- Horse riding